



**POST GRADUATE INSTITUTE OF SCIENCE
UNIVERSITY OF PERADENIYA**

DS 5216 - Artificial Intelligence

**Programming Assignment 02
Player Tracking in Sports Videos**

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Introduction

This project focuses on developing a computer vision–based system for player detection, tracking, and pose estimation in sports videos. The objective is to analyze short sports clips and automatically detect players, track their movements across frames, and estimate body keypoints using deep learning models.

Two main tasks were implemented:

- Player detection with tracking using a YOLOv8 model
- Pose estimation using a YOLOv8 pose model.

Dataset Description

The dataset consists of short sports video clips stored in Google Drive.

- Source: YouTube sports videos.
- Sports: Football, cricket, and rugby.
- Duration: 5–10 seconds per clip.
- Format: MP4 videos.
- Number of videos: 8.

The videos contain dynamic scenes with multiple moving players, making them suitable for real-world detection and tracking experiments.

Implementation

Environment Setup: The system was implemented in Google Colab using Python.

Required libraries such as OpenCV, PyTorch, and Ultralytics (YOLOv8) were installed.

GPU availability was checked to improve processing speed.

Google Drive was mounted to access input videos and store output results.

Player Detection and Tracking Model

A pretrained YOLOv8 nano model (yolov8n.pt) was used for player detection.

The model detects only the person class.

ByteTrack was used to assign unique IDs to players and track them across frames.

Each frame is processed to detect players, draw bounding boxes, and maintain tracking IDs.

Pose Estimation Model

A pretrained YOLOv8 pose estimation model (yolov8n-pose.pt) was used for keypoint detection. The model detects body joints such as head, shoulders, elbows, wrists, hips, knees, and ankles and connects them into a skeleton structure.

Output Processing

Each video was processed in two stages:

- Tracking output with bounding boxes and IDs, saved as tracked_video_i.mp4.
- Pose estimation output with skeleton visualization, saved as pose_video_i.mp4.

All outputs were stored in Google Drive.

Results and Analysis

The system performed well across football, cricket, and rugby videos.

Player detection worked effectively in crowded scenes.

Tracking was generally stable but showed minor identity switches in occlusion cases.

Pose estimation captured key movements such as running and bowling, but accuracy decreased for distant players.

Limitations and Improvements

Limitations include reduced accuracy for distant players, tracking instability in crowded scenes, and occlusion issues. Future improvements include using larger YOLO models, improved tracking algorithms like BoT-SORT, and temporal smoothing for pose estimation.

Conclusion

This project successfully demonstrates a deep learning-based system for player detection, tracking, and pose estimation in sports videos. The system can be extended for advanced sports analytics applications.